

JAVA SDE-2 INTERVIEW PREPARATION SERIES

DATA STRUCTURES AND ALGORITHMS - BOOK 17 OF 17 - STUDY STEP 17

Dynamic Programming: State, Transitions, and Optimization

Memoization, Tabulation, Reconstruction, Dimensions, and Space Compression

BY

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Derive dynamic programs from state meaning, transitions, base cases, evaluation order, and reconstruction instead of memorizing tables.

STATE | TRANSITION | MEMOIZATION | TABULATION | RECONSTRUCTION | DP

JAVA 21 | INTERVIEW CORE | SDE-2 FOLLOW-UPS | PRINTABLE EDITION

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Start Here - Your Route Through This Volume

60-second entry rule: If two or more recognition signals below expose a gap, begin with Chapter 1. If the prerequisites are weak, step back to **DSA 16 – Greedy Algorithms**. If you can already explain every outcome without notes, jump to Part II and prove it through the practice ladder.

This volume makes dynamic programming a specification process: define exactly what a state means, how it depends on smaller states, and what information can be compressed.

Readiness map

Decision	Evidence
START HERE	Subproblems overlap; A choice affects a reusable suffix or prefix state; Brute-force recursion repeats the same arguments; The problem asks for optimum, count, or feasibility under constraints.
PREPARE FIRST	Study Steps 01-16; Recursion, arrays, strings, and complexity.
MOVE ON WHEN	Define DP state and transitions precisely; Choose memoization or tabulation; Reconstruct a solution and compress safe dimensions; Analyze state count times transition cost.

Choose the route that fits today

- 1 **Learning:** read in order, type the representative Java examples, and complete each dry run.
- 2 **Revision or rehearsal:** start at the first decision map, invariant, or failure mode you cannot explain; if none expose a gap, go directly to Part II and prove readiness without notes.

DSA Segment Position

DSA 16 - Greedy Algorithms	YOU ARE HERE DSA 17 - Dynamic Programming	SEGMENT COMPLETE
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Part I - Core Learning**SDE-2 CORE CHAPTER**

Chapter 1 - Dynamic Programming Foundations: Derive State Before Writing a Table

Dynamic programming is a way to evaluate repeated subproblems once. It is not a collection of unrelated formulas. A correct DP starts with a precise state sentence.

See repetition in a recursive tree

For Fibonacci:

TEXT

```
fib(5)
  fib(4)
    fib(3)
      fib(2)
        fib(3)      repeated
```

The recursion has optimal substructure - larger answers combine smaller answers - and overlapping subproblems - the same arguments recur.

JAVA

```
static long fibonacciMemo(int n, long[] memo) {
    if (n <= 1) return n;
    if (memo[n] != -1L) return memo[n];
    memo[n] = fibonacciMemo(n - 1, memo) + fibonacciMemo(n - 2, memo);
    return memo[n];
}
```

Memoization preserves demand-driven recursion but still uses call-stack depth. Tabulation chooses an explicit evaluation order:

JAVA

```
static long fibonacci(int n) {
    if (n < 0) throw new IllegalArgumentException("negative n");
    if (n <= 1) return n;
    long previous = 0L;
    long current = 1L;
    for (int index = 2; index <= n; index++) {
        long next = Math.addExact(previous, current);
        previous = current;
        current = next;
    }
    return current;
}
```

Space compression is safe because each state needs only the previous two states. Numeric overflow remains a separate correctness boundary.

The six questions for every DP

- 1 **State:** What does `dp[...]` mean in one complete sentence?
- 2 **Choices:** What can the current decision do?
- 3 **Transition:** Which smaller states produce this state?
- 4 **Base:** Which states are known without recurrence?
- 5 **Order:** Are dependencies computed before use?
- 6 **Answer:** Which state or aggregate is returned?

Add two more for SDE-2:

- 7 **Reconstruction:** What parent/choice evidence recovers an actual solution?
- 8 **Compression:** Which older states are provably dead?

Example: House Robber from a state sentence

State:

TEXT

```
dp[i] = maximum amount obtainable from houses in indexes [0, i)
```

At boundary `i`, either skip house `i-1` and keep `dp[i-1]`, or take it and add its value to `dp[i-2]`.

TEXT

```
dp[i] = max(dp[i-1], dp[i-2] + value[i-1])
dp[0] = 0
dp[1] = max(0, value[0]) under an allow-skip-all contract
```

This formulation exposes boundary indexes and handles empty input naturally.

Memoization versus tabulation

Question	Memoization	Tabulation
Evaluation	only reached states	chosen order, often all states
Stack	recursive depth	usually none
Sparse state	can avoid unreachable states	may need explicit sparse map
Ordering	follows recursion	must be derived
Debugging	mirrors recurrence	table can be inspected

Neither is automatically faster. Compare reachable-state count, call overhead, iteration locality, and stack safety.

One-dimensional and two-dimensional state

Add a dimension only when the future answer depends on that information. For 0/1 knapsack:

TEXT

```
dp[item][capacity] = best value using first item items within capacity
```

The item dimension distinguishes which choices remain. The capacity dimension distinguishes remaining resource. The transition compares skipping and taking the current item.

Evaluation order is correctness

When compressing 0/1 knapsack to one dimension, capacities must iterate downward. Upward iteration would reuse the current item multiple times, silently changing the problem to unbounded knapsack.

JAVA

```
for each item:
    for capacity from limit down to itemWeight:
        dp[capacity] = max(dp[capacity], dp[capacity - itemWeight] + itemValue)
```

For unlimited reuse, upward iteration is often intentional. Loop direction encodes the reuse contract.

Counting, feasibility, and optimization differ

The same input shape can ask:

- feasibility: can a sum be formed? use boolean state;
- count: how many ways? use additive state and define order/combinations;
- minimum: fewest items? use infinity sentinel and min transition;
- maximum: best value? use max transition;
- reconstruction: which choices? store parents or retain enough table state.

Do not transplant a recurrence without restating the state.

Complexity from state space

TEXT

```
time = number of reachable states * transition work per state
space = stored states + recursion depth + reconstruction/output
```

For $n * \text{capacity}$ knapsack states with $O(1)$ transition work, time is $O(n * \text{capacity})$. This is pseudo-polynomial because capacity is a numeric value, not its bit-length.

Compression caution

Compression can destroy information needed to reconstruct choices, process dependencies in the wrong order, or obscure correctness. Derive a full correct table first, draw dependency arrows, then retain only live layers.

TRY IT | Foundation checkpoint

- 1 State House Robber DP in one sentence.
- 2 Why does memoization reduce repeated work but not recursion depth?
- 3 Why does 0/1 knapsack iterate capacity downward after compression?
- 4 What distinguishes combination count from permutation count?
- 5 How do you derive complexity without memorizing it?

SDE-2 CORE CHAPTER

Chapter 2 - SDE-2 Dynamic Programming: Derivation, Reconstruction, and Optimization

Why dynamic programming feels harder than it is

Dynamic programming (DP) is not a list of formulas. It is a method for organizing repeated subproblems when an optimum or count can be composed from smaller states. Most failures come from choosing an ambiguous state, omitting a base case, evaluating dependencies in the wrong order, or compressing storage before correctness is clear.

At SDE-2 level, derive the recurrence out loud, count states and transitions, identify numeric and memory limits, and reconstruct the requested choices rather than returning only a score. This chapter builds a repeatable derivation across one-dimensional decisions, grids, knapsack, subset and coin problems, subsequences, edit operations, stock state machines, and advanced interval/tree/bitmask families.

Learning objectives

After completing this chapter, you should be able to:

- distinguish overlapping subproblems from greedy-choice structure and brute force;
- derive state, transition, base cases, evaluation order, and answer location;
- move between memoization and tabulation;
- count complexity as number of reachable states times work per state;
- solve house robber, grid path, 0/1 knapsack, subset sum, and coin change;
- derive LIS, LCS, edit distance, and stock transaction states;
- reconstruct selected items or a sequence from a DP table;
- compress memory only when discarded states are no longer dependencies;
- recognize interval, tree, and bitmask DP boundaries; and
- discuss overflow, unreachable sentinels, recursion depth, allocation, and production scale.

The six-step derivation protocol

For every DP problem, write these six lines before code.

- 1 **State:** what exact subproblem does `dp[...]` answer?
- 2 **Choice/transition:** which first or last decision relates this state to smaller states?
- 3 **Base:** what are the smallest valid states and impossible states?
- 4 **Order:** in which order are all dependencies already known?
- 5 **Answer:** which state or aggregate is the requested result?
- 6 **Complexity:** how many states exist and how much work evaluates each?

Example for climbing n steps using moves of one or two:

TEXT

```
state: ways(i) = number of ways to reach exactly step i
transition: ways(i) = ways(i - 1) + ways(i - 2)
base: ways(0)=1, ways(negative)=0
order: increasing i, or recursive memoization
answer: ways(n)
states: n + 1, O(1) work each -> O(n) time
```

The empty construction base `ways(0)=1` is important: it provides one way to complete a recurrence exactly, not zero physical movements.

Recognition: DP versus alternatives

DP is promising when:

- a recursive search revisits the same parameter combination;
- the answer for a prefix, suffix, interval, capacity, or finite state can be composed from smaller ones;
- choices interact, so a local greedy ranking lacks an exchange proof;
- the input asks for an optimum, count, feasibility, or reconstruction; and
- the state space is small enough to enumerate.

DP is not automatically appropriate when states are unique, an invariant yields a linear greedy solution, or dimensions make the state space exponential. A memoized exponential state definition is still exponential.

Memoization versus tabulation

Top-down memoization

Write the recurrence as a recursive function and cache results by state. Benefits:

- follows the mathematical definition;
- computes only reachable states; and
- can be easier for irregular transitions.

Costs:

- recursion frames and possible `StackOverflowError`;
- hash/boxing overhead for non-array keys; and
- less predictable locality.

The memo must distinguish "uncomputed" from a legitimate result such as zero. Use a separate seen array, nullable boxed entries, or a sentinel outside the result domain.

Bottom-up tabulation

Allocate states and fill them in dependency order. Benefits:

- no recursion depth risk;
- predictable memory and often better locality; and
- easier rolling-array compression.

Costs:

- may compute unreachable states; and
- order can be less intuitive for interval or tree structures.

The toolkit implements climbing both ways. Both take $O(n)$ time. The tabulated version keeps only two predecessors and uses $O(1)$ space; memoization uses $O(n)$ memo and stack.

Pattern 1: house robber / nonadjacent selection

Given values along a line, select a maximum-sum subset with no adjacent indexes. Define:

TEXT

```
dp[i] = best value using prefix [0, i)
```

At last prefix item $i - 1$, either skip it and keep $dp[i - 1]$, or take it and add its value to $dp[i - 2]$:

TEXT

```
dp[i] = max(dp[i - 1], dp[i - 2] + value[i - 1])
dp[0] = 0
dp[1] = max(0, value[0])
```

Invariant while scanning: `previousOne` and `previousTwo` are optimal results for the two preceding prefix lengths. Only those states are dependencies, so storage compresses to $O(1)$.

For `[2, 7, 9, 3, 1]`, prefix optima are `0, 2, 7, 11, 11, 12`; choose indexes 0, 2, 4 for 12. If negative values are allowed and selecting nothing is legal, zero is a valid answer. If at least one item is required, bases and result change.

Time $O(n)$, space $O(1)$ for score only. Reconstructing indexes normally needs the full table or stored decisions.

Pattern 2: grid path DP

For minimum cost from top-left to bottom-right with moves only right and down:

TEXT

```
dp[row][col] = grid[row][col] + min(dp[row-1][col], dp[row][col-1])
```

The graph of legal moves is a DAG ordered by `row + col`. First cell is its own cost; first row can come only from the left; first column only from above.

For grid:

TEXT

```
1 3 1
1 5 1
4 2 1
```

minimum prefix rows become `[1, 4, 5]`, `[2, 7, 6]`, `[6, 8, 7]`, answer 7. One path is right, right, down, down.

Time is $O(\text{rows} * \text{cols})$. Score-only storage compresses to one row of $O(\text{cols})$: before updating `dp[col]` it means the value from above; after update it means current row, while `dp[col-1]` is left. Obstacles need an unreachable sentinel and guarded addition. Arbitrary four-direction movement creates cycles and calls for graph shortest-path algorithms, not this acyclic recurrence.

Pattern 3: 0/1 knapsack

Each item has positive weight and nonnegative value and may be taken at most once. With capacity C :

TEXT

```
dp[i][c] = best value using first i items with capacity c
```

Skip item $i-1$, or take it if its weight fits:

TEXT

```
dp[i][c] = max(dp[i-1][c],
               value[i-1] + dp[i-1][c-weight[i-1]])
```

Both branches depend on the previous item row, enforcing at-most-once use. Base row zero is all zero.

TRACE IT | Dry run and reconstruction

Weights $[2, 3, 4]$, values $[4, 5, 7]$, capacity 6. Best value is 11 from items 0 and 2. Starting at $dp[3][6]$, compare with $dp[2][6]$; the value differs, so item 2 was selected and capacity becomes 2. Compare $dp[2][2]$ with $dp[1][2]$; they match, so skip item 1. Item 0 then differs from the zero row, so select it.

State count is $(n+1)(capacity+1)$, each $O(1)$, yielding $O(nC)$ time and space. This is pseudo-polynomial: capacity is numeric magnitude, not input bit length.

The runnable full-table implementation rejects a table above an explicit cell budget and computes dimensions in `long` before any `+1` result reaches an array allocation. In particular, `capacity == Integer.MAX_VALUE` cannot become a negative column count through overflow. A production API should choose its budget from measured heap limits or use score-only compression when reconstruction is unnecessary.

For score only, compress to $O(C)$ and iterate capacities downward. Downward order ensures $dp[c-weight]$ still represents the previous item row. Iterating upward accidentally permits the same item repeatedly and solves an unbounded variant. Reconstruction becomes harder after compression; retain decisions, rerun portions, or accept full storage.

Pattern 4: subset sum

For nonnegative values, `possible[s]` says whether some processed subset totals s . Seed `possible[0]=true`. For each value, iterate s downward from target and set:

TEXT

```
possible[s] |= possible[s - value]
```

Invariant after an item: `possible[s]` represents subsets using only processed items, each at most once. Descending order prevents reading a state just created by the same item.

Time $O(n * \text{target})$, space $O(\text{target})$. Zero values do not change feasibility but matter for counting subsets. Negative values invalidate the simple $0..target$ index range; use an offset range, a set of reachable sums, meet-in-the-middle, or another constraint-specific method.

Equal-partition is subset sum with target $\text{total}/2$ after checking the total is even. Widen total before addition.

Pattern 5: coin change distinctions

"Coin change" names several different DPs.

Minimum number of coins, unlimited reuse

TEXT

```
dp[amount] = minimum coins to form amount
dp[0] = 0
dp[a] = 1 + min(dp[a - coin]) over fitting coins
```

Use an unreachable sentinel and never add one to it blindly. Loop amount increasing so smaller amounts are known. Time $O(\text{target} * \text{coinTypes})$, space $O(\text{target})$.

For coins $[1, 3, 4]$, target 6, minimum is two coins $3+3$; the locally largest coin 4 leaves 2 and produces $4+1+1$, showing why a generic largest-coin greedy rule fails.

Count combinations, unlimited reuse

Seed $\text{ways}[0]=1$; loop coins outside and amounts increasing inside. This counts each multiset once. Swapping loop order counts ordered sequences/permutations instead. For 0/1 use, amounts descend. Loop order is part of the problem semantics, not a micro-optimization.

Counts can grow far beyond `long`; define modulus or use `BigInteger` when exact large counts matter.

Pattern 6: longest increasing subsequence

Quadratic DP defines $\text{dp}[i]$ as LIS length ending exactly at i :

TEXT

```
dp[i] = 1 + max(dp[j]) for j < i and values[j] < values[i]
```

This takes $O(n^2)$ time and $O(n)$ space and reconstructs naturally with predecessor indexes.

The $O(n \log n)$ method maintains $\text{tails}[\text{length}-1]$, the smallest possible tail value of any increasing subsequence of that length seen so far. For each value, binary-search the first tail greater than or equal to it and replace that tail, or extend the array.

Invariant: tails are sorted, and a smaller tail is at least as extendable as a larger tail for the same length. Replacing a tail does not claim the tails array itself is a subsequence; it preserves existence of some subsequence for each length.

For `[10,9,2,5,3,7,101,18]`, tails evolve `[10]`, `[9]`, `[2]`, `[2,5]`, `[2,3]`, `[2,3,7]`, `[2,3,7,101]`, `[2,3,7,18]`; length 4. Strict LIS uses lower bound $\geq$; nondecreasing subsequence uses upper bound $>$.

Reconstructing the $O(n \log n)$ sequence requires predecessor and tail-index arrays, not just tail values.

Pattern 7: longest common subsequence

For code-point arrays `a` and `b`:

TEXT

```
dp[i][j] = LCS length of prefixes a[0..i) and b[0..j)
```

If last code points match, include them: $1 + dp[i-1][j-1]$. Otherwise skip one side: $\max(dp[i-1][j], dp[i][j-1])$. Empty prefix rows/columns are zero.

For `ABCBDBAB` and `BDCABA`, an LCS length is 4, such as `BCBA`. Reconstruction walks backward: matching symbols are chosen; otherwise move toward a neighboring state with the same optimal value. Ties permit multiple correct sequences, so deterministic tie policy matters.

Time and full reconstruction space are $O(nm)$. Score only compresses to $O(\min(n,m))$, but ordinary backtracking information is lost. Hirschberg's algorithm reconstructs with linear additional space through divide and conquer when needed.

Pattern 8: edit distance

Levenshtein distance permits insertion, deletion, and substitution at unit cost.

TEXT

```
dp[i][j] = minimum edits from first i code points to first j
base: dp[i][0]=i, dp[0][j]=j
if equal: dp[i][j]=dp[i-1][j-1]
else: 1 + min(delete dp[i-1][j],
               insert dp[i][j-1],
               replace dp[i-1][j-1])
```

Every transition identifies the last operation, so optimal substructure follows by removing that operation. State count $O(nm)$, transition work constant. The toolkit compresses to two rows. To reconstruct the edit script, retain the table or decisions.

Costs can be weighted, transposition can be added under a different recurrence, and Unicode normalization/unit choices belong to the contract. Code-point distance is not grapheme or linguistic distance.

Pattern 9: stock trading as a state machine

For at most k transactions, define:

- `cash[t]`: best profit after at most t completed sales while holding no stock;
- `hold[t]`: best profit after buying for transaction t and currently holding one stock.

For each price:

TEXT

```
hold[t] = max(hold[t], cash[t-1] - price)
cash[t] = max(cash[t], hold[t] + price)
```

Impossible hold states start at negative infinity. The implementation uses a safe sentinel so addition cannot overflow. State count is $O(k)$ per day; time $O(nk)$, space $O(k)$.

When $k \geq n/2$, the transaction cap cannot bind because each profitable transaction needs at least a buy and later sell day. Sum every positive adjacent increase in $O(n)$ time.

Cooldown, fees, multiple simultaneous holdings, and mandatory transactions change state and transitions. Write the state meaning first instead of patching conditionals onto a remembered formula.

Advanced pattern boundaries

Interval DP

State spans a contiguous interval, often `[left, right)`. Choose a split or last action inside it. Matrix-chain multiplication, optimal parenthesization, and burst balloons are examples. Fill shorter intervals before longer ones. Typical state count is $O(n^2)$ and trying every split yields $O(n^3)$ time.

Tree DP

Root the tree and compute states after children. House robber on a tree returns two values per node: best if this node is excluded and best if included. Included forces children excluded; excluded may choose each child's better state. Complexity is linear, but recursion depth may require an iterative postorder.

Bitmask DP

For small n , a mask represents a chosen subset. Traveling-salesperson-style state `dp[mask][last]` stores best cost to visit exactly `mask` and end at `last`. There are $n * 2^n$ states and up to n transitions each, $O(n^2 2^n)$ time. This is powerful around n in the teens or low twenties, not a polynomial escape from large input.

Digit DP

Count numbers satisfying a property with state such as (position, tight, started, summary). Memoization is valid only when the state includes every future-relevant constraint. This is a recognition boundary rather than a default interview tool.

Reconstruction and compression tension

A score is not always the requested answer. To reconstruct:

- store a parent/choice per state;
- compare a full state with the transition that could have produced it;
- retain predecessor indexes; or
- rerun selected subproblems when memory is more valuable than time.

Compression is safe only when future values no longer need discarded states. Iteration direction matters in knapsack. In-place grid rows rely on old-above/new-left meanings. Reconstruction may require the information compression removes. State this trade-off before claiming $O(1)$ space for a problem that must return a path or chosen set.

Runnable Java 21 reference implementation

Run with `java -ea DynamicProgrammingPatterns`.

JAVA (continued 1/11)

```
import java.util.ArrayList;
import java.util.Arrays;
import java.util.Collections;
import java.util.List;

public final class DynamicProgrammingPatterns {
    private static final long NEGATIVE_INFINITY = Long.MIN_VALUE / 4;
    private static final long MAX_KNAPSACK_CELLS = 20_000_000L;

    private DynamicProgrammingPatterns() {}

    public record KnapsackResult(long value, List<Integer> selectedIndexes) {}

    public static long climbWaysMemoized(int steps) {
        validateStepCount(steps);
        long[] memo = new long[steps + 1];
        Arrays.fill(memo, -1);
        return climb(steps, memo);
    }

    public static long climbWaysTabulated(int steps) {
        validateStepCount(steps);
        if (steps == 0) {
            return 1;
        }
        long twoBack = 1;
        long oneBack = 1;
        for (int step = 2; step <= steps; step++) {
            long current = Math.addExact(oneBack, twoBack);
            twoBack = oneBack;
            oneBack = current;
        }
    }
}
```

JAVA (continued 2/11)

```
        return oneBack;
    }

    public static long maxNonAdjacentSum(int[] values) {
        requireArray(values);
        long twoBack = 0;
        long oneBack = 0;
        for (int value : values) {
            long current = Math.max(oneBack, Math.addExact(twoBack, value));
            twoBack = oneBack;
            oneBack = current;
        }
        return oneBack;
    }

    public static long minimumGridPath(int[][] grid) {
        int cols = validateGrid(grid);
        long[] best = new long[cols];
        for (int row = 0; row < grid.length; row++) {
            for (int col = 0; col < cols; col++) {
                if (row == 0 && col == 0) {
                    best[col] = grid[row][col];
                } else if (row == 0) {
                    best[col] = Math.addExact(best[col - 1], grid[row][col]);
                } else if (col == 0) {
                    best[col] = Math.addExact(best[col], grid[row][col]);
                } else {
                    best[col] = Math.addExact(Math.min(best[col], best[col - 1]),
                        grid[row][col]);
                }
            }
        }
        return best[cols - 1];
    }
}
```

JAVA (continued 3/11)

```
public static KnapsackResult knapsack01(int[] weights, int[] values, int capacity) {
    if (weights == null || values == null || weights.length != values.length
        || capacity < 0) {
        throw new IllegalArgumentException("invalid knapsack input");
    }
    int n = weights.length;
    long rows = (long) n + 1;
    long columns = (long) capacity + 1;
    long cells = Math.multiplyExact(rows, columns);
    if (rows > Integer.MAX_VALUE || columns > Integer.MAX_VALUE
        || cells > MAX_KNAPSACK_CELLS) {
        throw new IllegalArgumentException("knapsack table exceeds allocation budget");
    }
    long[][] best = new long[n + 1][capacity + 1];
    for (int i = 1; i <= n; i++) {
        if (weights[i - 1] <= 0 || values[i - 1] < 0) {
            throw new IllegalArgumentException("positive weights and nonnegative values required");
        }
        for (int remaining = 0; remaining <= capacity; remaining++) {
            best[i][remaining] = best[i - 1][remaining];
            if (weights[i - 1] <= remaining) {
                best[i][remaining] = Math.max(best[i][remaining],
                    Math.addExact(values[i - 1],
                        best[i - 1][remaining - weights[i - 1]]));
            }
        }
    }
    List<Integer> selected = new ArrayList<>();
    int remaining = capacity;
    for (int i = n; i >= 1; i--) {
        if (best[i][remaining] != best[i - 1][remaining]) {
            selected.add(i - 1);
            remaining -= weights[i - 1];
        }
    }
}
```


JAVA (continued 4/11)

```
    }
}
Collections.reverse(selected);
return new KnapsackResult(best[n][capacity], List.copyOf(selected));
}

public static boolean subsetSum(int[] values, int target) {
    requireArray(values);
    if (target < 0 || target == Integer.MAX_VALUE) {
        throw new IllegalArgumentException("target must be in 0..Integer.MAX_VALUE-1");
    }
    boolean[] possible = new boolean[target + 1];
    possible[0] = true;
    for (int value : values) {
        if (value < 0) {
            throw new IllegalArgumentException("values must be nonnegative");
        }
        for (int sum = target; sum >= value; sum--) {
            possible[sum] |= possible[sum - value];
        }
    }
    return possible[target];
}

public static int minimumCoins(int[] coins, int target) {
    requireArray(coins);
    if (target < 0 || target == Integer.MAX_VALUE) {
        throw new IllegalArgumentException("invalid target");
    }
    for (int coin : coins) {
        if (coin <= 0) {
            throw new IllegalArgumentException("coins must be positive");
        }
    }
}
```

JAVA (continued 5/11)

```
    }
    int unreachable = target + 1;
    int[] best = new int[target + 1];
    Arrays.fill(best, unreachable);
    best[0] = 0;
    for (int amount = 1; amount <= target; amount++) {
        for (int coin : coins) {
            if (coin <= amount && best[amount - coin] != unreachable) {
                best[amount] = Math.min(best[amount], best[amount - coin] + 1);
            }
        }
    }
    return best[target] == unreachable ? -1 : best[target];
}

public static int lisLength(int[] values) {
    requireArray(values);
    int[] tails = new int[values.length];
    int size = 0;
    for (int value : values) {
        int low = 0;
        int high = size;
        while (low < high) {
            int middle = low + (high - low) / 2;
            if (tails[middle] < value) {
                low = middle + 1;
            } else {
                high = middle;
            }
        }
        tails[low] = value;
        if (low == size) {
            size++;
        }
    }
}
```

JAVA (continued 6/11)

```
    }  
    }  
    return size;  
}  
  
public static String longestCommonSubsequence(String first, String second) {  
    requireString(first);  
    requireString(second);  
    int[] a = first.codePoints().toArray();  
    int[] b = second.codePoints().toArray();  
    int[][] length = new int[a.length + 1][b.length + 1];  
    for (int i = 1; i <= a.length; i++) {  
        for (int j = 1; j <= b.length; j++) {  
            if (a[i - 1] == b[j - 1]) {  
                length[i][j] = length[i - 1][j - 1] + 1;  
            } else {  
                length[i][j] = Math.max(length[i - 1][j], length[i][j - 1]);  
            }  
        }  
    }  
    List<Integer> reversed = new ArrayList<>();  
    int i = a.length;  
    int j = b.length;  
    while (i > 0 && j > 0) {  
        if (a[i - 1] == b[j - 1]) {  
            reversed.add(a[i - 1]);  
            i--;  
            j--;  
        } else if (length[i - 1][j] >= length[i][j - 1]) {  
            i--;  
        } else {  
            j--;  
        }  
    }  
}
```

JAVA (continued 7/11)

```
    }
    Collections.reverse(reversed);
    StringBuilder result = new StringBuilder();
    for (int point : reversed) {
        result.appendCodePoint(point);
    }
    return result.toString();
}

public static int editDistance(String first, String second) {
    requireString(first);
    requireString(second);
    int[] a = first.codePoints().toArray();
    int[] b = second.codePoints().toArray();
    if (b.length > a.length) {
        int[] temporary = a;
        a = b;
        b = temporary;
    }
    int[] previous = new int[b.length + 1];
    int[] current = new int[b.length + 1];
    for (int j = 0; j <= b.length; j++) {
        previous[j] = j;
    }
    for (int i = 1; i <= a.length; i++) {
        current[0] = i;
        for (int j = 1; j <= b.length; j++) {
            if (a[i - 1] == b[j - 1]) {
                current[j] = previous[j - 1];
            } else {
                current[j] = 1 + Math.min(previous[j - 1],
                    Math.min(previous[j], current[j - 1]));
            }
        }
    }
}
```

JAVA (continued 8/11)

```
    }
    int[] temporary = previous;
    previous = current;
    current = temporary;
}
return previous[b.length];
}

public static long maxStockProfit(int[] prices, int transactions) {
    requireArray(prices);
    if (transactions < 0) {
        throw new IllegalArgumentException("transactions must be nonnegative");
    }
    for (int price : prices) {
        if (price < 0) {
            throw new IllegalArgumentException("prices must be nonnegative");
        }
    }
    if (transactions == 0 || prices.length < 2) {
        return 0;
    }
    if (transactions >= prices.length / 2) {
        long profit = 0;
        for (int day = 1; day < prices.length; day++) {
            if (prices[day] > prices[day - 1]) {
                profit += (long) prices[day] - prices[day - 1];
            }
        }
        return profit;
    }
    long[] cash = new long[transactions + 1];
    long[] hold = new long[transactions + 1];
    Arrays.fill(hold, NEGATIVE_INFINITY);
```

JAVA (continued 9/11)

```
        for (int price : prices) {
            for (int completed = 1; completed <= transactions; completed++) {
                hold[completed] = Math.max(hold[completed], cash[completed - 1] - price);
                cash[completed] = Math.max(cash[completed], hold[completed] + price);
            }
        }
        return cash[transactions];
    }

    private static long climb(int steps, long[] memo) {
        if (steps <= 1) {
            return 1;
        }
        if (memo[steps] != -1) {
            return memo[steps];
        }
        memo[steps] = Math.addExact(climb(steps - 1, memo), climb(steps - 2, memo));
        return memo[steps];
    }

    private static void validateStepCount(int steps) {
        if (steps < 0 || steps > 91) {
            throw new IllegalArgumentException("steps must be in 0..91 for long output");
        }
    }

    private static int validateGrid(int[][] grid) {
        if (grid == null || grid.length == 0 || grid[0] == null
            || grid[0].length == 0) {
            throw new IllegalArgumentException("nonempty grid required");
        }
        int cols = grid[0].length;
        for (int[] row : grid) {
```

JAVA (continued 10/11)

```
        if (row == null || row.length != cols) {
            throw new IllegalArgumentException("grid must be rectangular");
        }
    }
    return cols;
}

private static void requireArray(int[] values) {
    if (values == null) {
        throw new IllegalArgumentException("array must not be null");
    }
}

private static void requireString(String value) {
    if (value == null) {
        throw new IllegalArgumentException("string must not be null");
    }
}

public static void main(String[] args) {
    assert climbWaysMemoized(10) == 89;
    assert climbWaysTabulated(10) == 89;
    assert maxNonAdjacentSum(new int[] {2, 7, 9, 3, 1}) == 12;
    assert maxNonAdjacentSum(new int[] {-4, -2}) == 0;
    assert minimumGridPath(new int[][] {{1, 3, 1}, {1, 5, 1}, {4, 2, 1}}) == 7;

    KnapsackResult packed = knapsack01(new int[] {2, 3, 4},
        new int[] {4, 5, 7}, 6);
    assert packed.value() == 11;
    assert packed.selectedIndexes().equals(List.of(0, 2));
    boolean hugeCapacityRejected = false;
    try {
        knapsack01(new int[0], new int[0], Integer.MAX_VALUE);
    }
```

JAVA (continued 11/11)

```

    } catch (IllegalArgumentException expected) {
        hugeCapacityRejected = true;
    }
    assert hugeCapacityRejected;
    assert subsetSum(new int[] {3, 34, 4, 12, 5, 2}, 9);
    assert !subsetSum(new int[] {3, 34, 4, 12, 5, 2}, 30);
    assert minimumCoins(new int[] {1, 3, 4}, 6) == 2;
    boolean invalidCoinRejected = false;
    try {
        minimumCoins(new int[] {0}, 0);
    } catch (IllegalArgumentException expected) {
        invalidCoinRejected = true;
    }
    assert invalidCoinRejected;

    assert lisLength(new int[] {10, 9, 2, 5, 3, 7, 101, 18}) == 4;
    String lcs = longestCommonSubsequence("ABCBDAAB", "BDCABA");
    assert lcs.length() == 4;
    assert isSubsequence(lcs, "ABCBDAAB") && isSubsequence(lcs, "BDCABA");
    assert editDistance("kitten", "sitting") == 3;
    assert maxStockProfit(new int[] {3, 2, 6, 5, 0, 3}, 2) == 7;
}

private static boolean isSubsequence(String candidate, String source) {
    int index = 0;
    for (int point : source.codePoints().toArray()) {
        if (index < candidate.length() && candidate.codePointAt(index) == point) {
            index += Character.charCount(point);
        }
    }
    return index == candidate.length();
}
}

```

Complexity and state table

Family	States	Work/state	Time	Compressed space
climb/house robber	$O(n)$	$O(1)$	$O(n)$	$O(1)$
grid right/down	$O(\text{rows} \times \text{cols})$	$O(1)$	$O(\text{rows} \times \text{cols})$	$O(\text{cols})$
0/1 knapsack	$O(nC)$	$O(1)$	$O(nC)$	$O(C)$ score only
subset sum	$O(nT)$ conceptual	$O(1)$	$O(nT)$	$O(T)$
minimum coin change	$O(T)$	$O(\text{coins})$	$O(T \times \text{coins})$	$O(T)$
LIS quadratic	$O(n)$ states	$O(n)$	$O(n^2)$	$O(n)$
LIS tails	$O(n)$	$O(\log n)$ search	$O(n \log n)$	$O(n)$
LCS/edit distance	$O(nm)$	$O(1)$	$O(nm)$	$O(\min(n, m))$ score
stock at most k	$O(nk)$	$O(1)$	$O(nk)$	$O(k)$
interval split DP	$O(n^2)$	often $O(n)$	often $O(n^3)$	problem-specific
bitmask last-state	$O(2^n)$	up to $O(n)$	$O(2^n)$	exponential

WATCH OUT | Edge cases and common mistakes

- 1 **State lacks future information.** If two histories with the same key permit different futures, the state is incomplete.
- 2 **Legitimate zero used as uncomputed.** Separate memo status from values.
- 3 **Base for empty construction wrong.** Counts often use one empty way; maxima may use zero or negative infinity.
- 4 **Impossible state added blindly.** Guard sentinels before arithmetic.
- 5 **Iteration direction changes item reuse.** Descending capacity is 0/1; ascending enables unbounded reuse.
- 6 **Combination and sequence counts confused.** Coin/amount loop order changes meaning.
- 7 **All-negative selection contract.** Decide whether choosing nothing is legal.
- 8 **Grid recurrence used on cyclic moves.** Use graph algorithms when dependencies are not acyclic.
- 9 **LIS strictness wrong.** Lower-bound versus upper-bound determines duplicate handling.
- 10 **Compressed score claimed to reconstruct choices.** Preserve decisions or explain recomputation.
- 11 **Pseudo-polynomial called polynomial in input length.** Capacity/target magnitude matters.
- 12 **Count overflow.** Combinatorial counts exceed `long` quickly.
- 13 **Recursive memo stack ignored.** State count does not bound call depth safely for Java.
- 14 **Huge table allocated from untrusted input.** Validate product before allocation.
- 15 **Text unit unspecified.** Code points, UTF-16 units, and graphemes produce different sequence states.

SDE-2 production follow-ups

- **Allocation guards:** compute table dimensions in `long`, enforce budgets, and fail before allocating. $O(nm)$ may be impossible even when time sounds acceptable.
- **Sparse states:** memo maps can save memory when few states are reachable, but hashing and boxing add cost and cardinality still needs a bound.
- **Rolling storage:** reuse primitive arrays for locality and less GC pressure; clear only required ranges.
- **Overflow contract:** use exact arithmetic, modular counts, saturation, `BigInteger`, or explicit rejection. Silent wraparound destroys optimal comparisons.
- **Reconstruction audit:** return named choices and validate their score independently. This catches tie/backtracking bugs.
- **Incremental changes:** DP tables generally become stale when inputs change. Determine whether localized recomputation is valid or rebuild from a versioned snapshot.
- **Parallelism:** wavefront/grid phases or independent states may parallelize, but dependencies, synchronization, and memory bandwidth can dominate.
- **Approximation:** capacity or state explosion may require scaling, pruning, beam search, or approximation schemes. Label loss of exactness.
- **Persistence:** checkpointing large tables can cost more than recomputation; measure serialization and version recurrence semantics.
- **Observability:** record state count, reachable-state ratio, table bytes, cache hit rate, and reconstruction validation, not raw sensitive inputs.

TRY IT | Exercises with model checkpoints

Exercise 1: reconstruct house robber indexes

Return one optimal nonadjacent index set.

Model checkpoints: retain full prefix scores or decisions; backtrack by comparing skip with take; define ties deterministically; verify no adjacent indexes and recompute sum.

Exercise 2: grid with obstacles and path

Return minimum cost and coordinates.

Model checkpoints: unreachable sentinel; never add to sentinel; store parent direction or compare table predecessors; start/end blocked behavior; $O(\text{rows} \times \text{cols})$ output-aware memory.

Exercise 3: count coin combinations

Count ways to form target with unlimited coins.

Model checkpoints: validate distinct positive denominations or define duplicate treatment; coins outer, amount ascending; `ways[0]=1`; choose `BigInteger` or modulus; contrast permutation loop order.

Exercise 4: edit script

Return insert/delete/replace operations, not only distance.

Model checkpoints: full table or divide-and-conquer reconstruction; define tie order; indexes shift as edits apply, so represent operations against prefixes or apply from a safe direction; verify script transforms input.

Exercise 5: LIS reconstruction

Extend the $O(n \log n)$ method to return indexes.

Model checkpoints: `tailIndex[length]` stores input index; predecessor of current is previous length's tail index; final chain backtracks; tails values alone are not a sequence; strict duplicate policy.

Exercise 6: tree independent set

Maximum-weight set of nonadjacent tree vertices.

Model checkpoints: per node return (`excluded`, `included`); included adds excluded children, excluded adds max child states; postorder; parent prevents traversal back edge; iterative plan for deep trees.

Exercise 7: traveling salesperson bitmask DP

Find minimum Hamiltonian tour for small n .

Model checkpoints: state (`mask`, `last`) includes start; transition from previous endpoint; unreachable sentinel; close tour to start; $O(n^2 \cdot 2^n)$ time and $O(n \cdot 2^n)$ space; reject large n before shifting/allocating.

Interview answer checklist

- ☐ I wrote a one-sentence state meaning with exact indexes/ranges.
- ☐ I derived every transition from a last or first choice.
- ☐ I covered empty, impossible, and smallest bases.
- ☐ My evaluation order satisfies every dependency.
- ☐ I identified the exact answer state.
- ☐ I counted states times transition work.
- ☐ I distinguished pseudo-polynomial and exponential complexity.
- ☐ I compressed only after proving which history is unnecessary.
- ☐ I preserved enough information for requested reconstruction.
- ☐ I defined overflow, text units, and allocation limits.
- ☐ I can compare memoization, tabulation, greedy, graph, and search alternatives.

RECAP | Summary

Dynamic programming becomes systematic when every solution follows state, transition, base, order, answer, and complexity. Memoization mirrors recursion; tabulation exposes dependency order and compression. House robber retains two prefix states; grid paths follow a DAG; 0/1 knapsack and subset sum depend on descending capacity; coin loop order encodes reuse and ordering semantics. LIS has quadratic and tails-based formulations. LCS and edit distance align two sequences, while stock trading is a finite state machine over days and transaction counts. Interval, tree, and bitmask DP extend the same method. The SDE-2 distinction is honest state counting, reconstructable outputs, safe sentinels and arithmetic, and memory-aware production boundaries.

SDE-2 CORE CHAPTER

Chapter 3 - Essential Dynamic-Programming Pattern Clinics

The core chapter covers linear, grid, knapsack, sequence, edit, and stock state. Word break makes prefix feasibility concrete, while matrix-chain multiplication turns the interval-DP boundary into a complete derivation.

Clinic 1: word break as prefix feasibility

Define:

TEXT

```
reachable[end] = text[0..end) can be segmented into dictionary words
```

Base case `reachable[0] = true` represents the empty prefix. For every reachable start, try word endings up to the maximum dictionary word length. A matching word makes that end reachable.

This is not plain greedy. Choosing the longest or shortest matching word can block a valid later segmentation. DP retains all reachable prefix boundaries without materializing every complete sentence.

If n is the text length and L is the maximum dictionary word length, the state/transition structure is $O(nL)$, excluding the cost of substring creation and hashing. Java substring creates a new string in current mainstream JDKs, so a production implementation may use a trie, region comparison, or indexes to control allocation.

Clinic 2: matrix-chain interval DP

Matrices are described by dimensions `d[0..n]`; matrix i has shape `d[i] × d[i+1]`. Multiplication order changes scalar work but not the final matrix.

State:

TEXT

```
cost[left][right] = minimum scalar multiplications for matrices left through right
```

Base: one matrix needs zero multiplications. For every split between `left` and `right`:

TEXT

```
cost[left][split]
+ cost[split + 1][right]
+ d[left] * d[split + 1] * d[right + 1]
```

Evaluate shorter intervals before longer intervals. There are $O(n^2)$ states and $O(n)$ split choices per state, so time is $O(n^3)$ and table space is $O(n^2)$.

For dimensions `[40, 20, 30, 10, 30]`, the minimum cost is 26,000. Greedily multiplying the currently cheapest adjacent pair is not generally safe because each merge changes future dimensions.

Runnable Java 21 clinic

JAVA (continued 1/3)

```
import java.util.HashSet;
import java.util.Objects;
import java.util.Set;

public final class DynamicProgrammingCoverageClinic {
    private DynamicProgrammingCoverageClinic() {}

    public static boolean canSegment(String text, Set<String> dictionary) {
        Objects.requireNonNull(text, "text");
        Objects.requireNonNull(dictionary, "dictionary");
        Set<String> words = new HashSet<>();
        int maxLength = 0;
        for (String word : dictionary) {
            if (word == null || word.isEmpty()) {
                throw new IllegalArgumentException("dictionary words must be nonempty");
            }
            words.add(word);
            maxLength = Math.max(maxLength, word.length());
        }

        boolean[] reachable = new boolean[text.length() + 1];
        reachable[0] = true;
        for (int start = 0; start < text.length(); start++) {
            if (!reachable[start]) {
                continue;
            }
            for (String word : words) {
                int end = start + word.length();
                if (end > text.length()) {
                    continue;
                }
                if (text.substring(start, end).equals(word)) {
                    reachable[end] = true;
                }
            }
        }
        return reachable[text.length()];
    }
}
```

JAVA (continued 2/3)

```

    }
    int lastEnd = Math.min(text.length(), start + maxLength);
    for (int end = start + 1; end <= lastEnd; end++) {
        if (words.contains(text.substring(start, end))) {
            reachable[end] = true;
        }
    }
}
return reachable[text.length()];
}

public static long minimumMatrixChainCost(int[] dimensions) {
    Objects.requireNonNull(dimensions, "dimensions");
    if (dimensions.length < 2) {
        throw new IllegalArgumentException("at least one matrix is required");
    }
    for (int dimension : dimensions) {
        if (dimension <= 0) {
            throw new IllegalArgumentException("dimensions must be positive");
        }
    }

    int matrices = dimensions.length - 1;
    long[][] cost = new long[matrices][matrices];
    for (int length = 2; length <= matrices; length++) {

```

JAVA (continued 3/3)

```

        for (int left = 0; left + length <= matrices; left++) {
            int right = left + length - 1;
            cost[left][right] = Long.MAX_VALUE;
            for (int split = left; split < right; split++) {
                long multiply = Math.multiplyExact(
                    Math.multiplyExact((long) dimensions[left],
                        dimensions[split + 1]),
                    dimensions[right + 1]);
                long candidate = Math.addExact(
                    Math.addExact(cost[left][split], cost[split + 1][right]),
                    multiply);
                cost[left][right] = Math.min(cost[left][right], candidate);
            }
        }
    }
    return cost[0][matrices - 1];
}

public static void main(String[] args) {
    assert canSegment("leetcode", Set.of("leet", "code"));
    assert !canSegment("catsanddog", Set.of("cats", "dog", "sand", "and", "cat"));
    assert minimumMatrixChainCost(new int[] {40, 20, 30, 10, 30}) == 26_000;
    System.out.println("PASS essential dynamic-programming clinics");
}
}

```

Expected output with assertions enabled:

TEXT

PASS essential dynamic-programming clinics

Interviewer follow-up chain with model answers

Interviewer: How would you return one word-break segmentation?

Candidate: Store the predecessor start whenever an end first becomes reachable. If the final index is reachable, walk predecessors backward and reverse the words. If a specific optimal segmentation is required, the state must encode that objective.

Interviewer: Why is matrix-chain DP evaluated by increasing interval length?

Candidate: Every transition reads two strict subintervals. Increasing length is a topological order of that dependency graph.

Interviewer: Can matrix-chain storage be compressed to one dimension?

Candidate: Not by the simple rolling technique used for linear DP. Many nonadjacent subinterval states remain live for larger intervals. If only the cost is needed, specialized optimizations require additional mathematical structure; they are not a generic compression rule.

SDE-2 CORE CHAPTER

Chapter 4 - Realistic Dynamic Programming Interview Rounds

Round 1: minimum coins for an amount

Prompt

Given positive coin denominations with unlimited reuse, return the fewest coins needed to form **amount**, or -1.

Candidate derivation

State: $dp[a]$ is the minimum coins required to form exact amount **a**; unreachable states use an infinity sentinel. Base $dp[0] = 0$. For each amount, try every coin that can be the final coin.

JAVA

```
static int minimumCoins(int[] coins, int amount) {
    if (amount < 0) throw new IllegalArgumentException("negative amount");
    int unreachable = amount + 1;
    int[] dp = new int[amount + 1];
    Arrays.fill(dp, unreachable);
    dp[0] = 0;
    for (int current = 1; current <= amount; current++) {
        for (int coin : coins) {
            if (coin <= 0) throw new IllegalArgumentException("positive coins required");
            if (coin <= current && dp[current - coin] != unreachable) {
                dp[current] = Math.min(dp[current], dp[current - coin] + 1);
            }
        }
    }
    return dp[amount] == unreachable ? -1 : dp[amount];
}
```

Follow-up answers

Why $amount + 1$ as infinity? With positive integer coins, any feasible solution uses at most **amount** coins when denomination 1 exists, and the sentinel avoids overflow from `Integer.MAX_VALUE + 1`. Without coin 1, it remains safely above every possible coin count.

Need the actual coins? Store the last chosen coin for each improved amount and walk backward. Define tie-breaking among equal-length solutions.

Complexity? $O(\text{amount} * \text{numberOfCoins})$ time and $O(\text{amount})$ space; pseudo-polynomial in numeric amount.

Round 2: longest common subsequence

Prompt

Return the length of the longest sequence appearing in order, not necessarily contiguously, in both strings.

State and transition

TEXT

$dp[i][j]$ = LCS length of first i chars of first and first j chars of second

If final characters match, extend $dp[i-1][j-1]$. Otherwise discard one final character and keep the better of $dp[i-1][j]$ and $dp[i][j-1]$.

JAVA

```
static int lcsLength(String first, String second) {
    int[][] dp = new int[first.length() + 1][second.length() + 1];
    for (int i = 1; i <= first.length(); i++) {
        for (int j = 1; j <= second.length(); j++) {
            if (first.charAt(i - 1) == second.charAt(j - 1)) {
                dp[i][j] = dp[i - 1][j - 1] + 1;
            } else {
                dp[i][j] = Math.max(dp[i - 1][j], dp[i][j - 1]);
            }
        }
    }
    return dp[first.length()][second.length()];
}
```

Follow-up answers

Substring instead? Contiguous matching resets on mismatch; the state and answer aggregation differ.

Space compression? Length needs only previous and current rows: $O(\min(m,n))$ space after choosing the shorter string as columns. Reconstruction generally needs the full table or a divide-and-conquer method.

Unicode? `charAt` compares UTF-16 code units. If the contract is code-point sequences, convert accordingly; user-perceived grapheme sequences need a stronger text boundary.

Round 3: equal-subset partition

Prompt

Return whether positive integers can be partitioned into two groups with equal sum.

Candidate derivation

The total must be even. The problem becomes 0/1 subset sum for target `total / 2`.

JAVA

```
static boolean canPartition(int[] values) {
    long total = 0L;
    for (int value : values) {
        if (value < 0) throw new IllegalArgumentException("nonnegative values required");
        total += value;
    }
    if ((total & 1L) != 0L || total / 2L > Integer.MAX_VALUE) return false;
    int target = (int) (total / 2L);
    boolean[] possible = new boolean[target + 1];
    possible[0] = true;
    for (int value : values) {
        for (int sum = target; sum >= value; sum--) {
            possible[sum] |= possible[sum - value];
        }
    }
    return possible[target];
}
```

Follow-up answers

Why downward? Each value may be used once. Upward iteration could read a state written earlier in the same item iteration and reuse that item.

Large values? $O(n * \text{target})$ may be infeasible even for moderate n . A meet-in-the-middle approach can suit small n with huge values; a bitset can improve constants for bounded sums; approximation depends on the business contract.

Closing answer pattern

Say the state sentence, choices, recurrence, bases, evaluation order, answer location, state-count complexity, numeric sentinel policy, reconstruction plan, and whether compression preserves required information.

SDE-2 CORE CHAPTER

Chapter 5 - Dynamic Programming Practice Lab

- 1 Draw repeated states in naive Fibonacci recursion.
- 2 State House Robber using a half-open prefix.
- 3 Compare memoization and tabulation for sparse reachable states.
- 4 Debug a memo array whose zero value means both "not computed" and a valid answer.
- 5 Debug 0/1 knapsack compressed with upward capacity iteration.
- 6 Count grid paths with obstacles and define numeric overflow policy.
- 7 Return one longest increasing subsequence, not only its length.
- 8 Distinguish coin-change combinations from permutations by loop order.
- 9 Reconstruct an edit script from edit-distance state.
- 10 Derive stock-trading state as day and holding status.
- 11 Define interval DP for matrix-chain multiplication or burst balloons.
- 12 Define tree DP for maximum nonadjacent node sum.
- 13 Explain pseudo-polynomial complexity.
- 14 Decide when space compression is worth losing reconstruction data.
- 15 Compare DP, greedy, and shortest-path formulations of the same state graph.
- 16 **Interview Core:** Solve word break as prefix feasibility, return one valid segmentation, and discuss substring-allocation cost in Java.

SDE-2 CORE CHAPTER

Chapter 6 - Dynamic Programming Practice Lab Solutions

- 1 `fib(n-2)` appears directly and again inside `fib(n-1)`; the recurrence tree repeats many argument values.
- 2 `dp[i]` is the maximum amount from houses `[0, i)`. Transition is max of skipping house `i-1` and taking it plus `dp[i-2]`.
- 3 Memoization can avoid unreachable states but retains recursion overhead/depth. Tabulation has explicit order and good locality but may evaluate every table cell.
- 4 Use a separate visited array, nullable wrappers, or a sentinel outside the valid answer range. Do not confuse a legitimate zero with missing state.
- 5 Iterate capacity downward so each item reads only states from the previous conceptual item layer. Upward iteration permits reuse.
- 6 `dp[row][column]` counts paths to that cell; obstacles contribute zero, and other cells add top plus left. Use long, exact arithmetic, modulo, or big integers according to the contract.
- 7 $O(n^2)$ DP stores predecessor index when a better length ends at `i`, then follows predecessors from the best endpoint. $O(n \log n)$ reconstruction tracks tail indexes and predecessors.
- 8 Coins outer and amounts upward count combinations without order. Amount outer and coins inner count ordered sequences under the corresponding recurrence.
- 9 Walk backward from `(m, n)`: a diagonal match emits nothing, diagonal mismatch emits replace, upward emits delete, and left emits insert. Tie-breaking defines which optimal script is returned.
- 10 `dp[day][holding]` is best profit from a day under current holding status. Transitions compare rest with buy/sell, extended for transaction count, fee, or cooldown.
- 11 State over interval `[left, right]`; try every final split/burst and combine independent subinterval answers plus boundary cost. Evaluate by increasing interval length.
- 12 Each node returns two values: best when node is taken and when skipped. Taking forces children skipped; skipping allows the better child state independently.
- 13 $O(n * \text{amount})$ is polynomial in numeric amount but exponential in the number of bits required to encode a potentially huge amount.
- 14 Compress only after dependency analysis. Keep the full table or parent choices when an actual solution path is required and memory permits.
- 15 DP evaluates a DAG of states, greedy commits using a proof, and shortest path optimizes edge-cost accumulation. Equivalent state graphs can reveal alternative algorithms, but edge weights and acyclicity determine validity.

- 16 Let `reachable[end]` mean prefix `[0,end)` can be segmented, with `reachable[0]=true`. From every reachable start, try dictionary words or endings up to the maximum word length; record the predecessor start on the first successful transition to each end. Reconstruct backward from `n` and reverse. The abstract transition count can be $O(nL)$, but Java substring creation and hashing add character/allocation cost; a trie or region comparison can avoid temporary strings.

SDE-2 CORE CHAPTER

Chapter 7 - Executable Dynamic-Programming Interview Checks

This dependency-free Java companion validates minimum-coin optimization, LCS state transitions, and downward 0/1 subset-sum compression. A successful run prints PASS 5 dynamic-programming checks.

JAVA (continued 1/2)

```
import java.util.Arrays;

public final class DynamicProgrammingInterviewChecks {
    private DynamicProgrammingInterviewChecks() {}

    static int minimumCoins(int[] coins, int amount) {
        if (amount < 0) throw new IllegalArgumentException("negative amount");
        int unreachable = amount + 1;
        int[] dp = new int[amount + 1];
        Arrays.fill(dp, unreachable);
        dp[0] = 0;
        for (int current = 1; current <= amount; current++) {
            for (int coin : coins) {
                if (coin <= 0) throw new IllegalArgumentException("positive coins required");
                if (coin <= current && dp[current - coin] != unreachable) {
                    dp[current] = Math.min(dp[current], dp[current - coin] + 1);
                }
            }
        }
        return dp[amount] == unreachable ? -1 : dp[amount];
    }

    static int lcsLength(String first, String second) {
        int[][] dp = new int[first.length() + 1][second.length() + 1];
        for (int i = 1; i <= first.length(); i++) {
            for (int j = 1; j <= second.length(); j++) {
                dp[i][j] = first.charAt(i - 1) == second.charAt(j - 1)
                    ? dp[i - 1][j - 1] + 1
                    : Math.max(dp[i - 1][j], dp[i][j - 1]);
            }
        }
        return dp[first.length()][second.length()];
    }
}
```

JAVA (continued 2/2)

```
static boolean canPartition(int[] values) {
    long total = 0L;
    for (int value : values) {
        if (value < 0) throw new IllegalArgumentException("nonnegative values required");
        total += value;
    }
    if ((total & 1L) != 0L || total / 2L > Integer.MAX_VALUE) return false;
    int target = (int) (total / 2L);
    boolean[] possible = new boolean[target + 1];
    possible[0] = true;
    for (int value : values) {
        for (int sum = target; sum >= value; sum--) {
            possible[sum] |= possible[sum - value];
        }
    }
    return possible[target];
}

private static void check(boolean condition, String message) {
    if (!condition) throw new AssertionError(message);
}

public static void main(String[] args) {
    check(minimumCoins(new int[] {1, 2, 5}, 11) == 3, "minimum coins");
    check(minimumCoins(new int[] {2}, 3) == -1, "unreachable");
    check(lcsLength("abcde", "ace") == 3, "LCS");
    check(canPartition(new int[] {1, 5, 11, 5}), "partition true");
    check(!canPartition(new int[] {1, 2, 3, 5}), "partition false");
    System.out.println("PASS 5 dynamic-programming checks");
}
}
```


Part II - Practice and Handoff

Practice Ladder and Next Steps

TRY IT | Practice ladder

- Foundation: Fibonacci, stairs, and one-dimensional choices
- Interview Core: grids, knapsack, subsequences, and string DP
- SDE-2 Follow-up: reconstruct decisions and control memory
- Challenge: interval, tree, or bitmask state

For every coding problem, state the input contract, select the numeric and collection types deliberately, write the invariant before the loop or recursion, test the smallest and largest valid inputs, and close with time and auxiliary-space complexity.

Navigation

[Previous book: DSA 16 - Greedy Algorithms: Recognition, Proofs, and Scheduling](#)

[Series index](#)

[Return to the complete series index](#)

TRY IT | Completion check

- ☐ I can recognize the core patterns without being told the data structure or algorithm.
- ☐ I can implement the representative Java templates from memory.
- ☐ I can explain why each implementation is correct.
- ☐ I can test boundaries, invalid inputs, overflow, and adversarial shapes.
- ☐ I can answer at least one SDE-2 follow-up involving trade-offs or production constraints.

Series Roadmap

Choose Java, DSA, or System Design and Backend first, then follow the books inside that segment in order. Stable study-step codes remain on filenames and links; the clearer segment book codes appear on the website and every PDF cover.

SEGMENT	BOOK	FOCUSED LEARNING PATH
Java	JAVA 01	Java Foundations for Problem Solving
Java	JAVA 02	Git and GitHub for Java Engineers
Java	JAVA 03	Maven and Gradle for Java Engineers
Java	JAVA 04	Advanced Java B: Language, OOP, and Modern Java
Java	JAVA 05	Advanced Java C: Collections, Streams, and I/O
Java	JAVA 06	Advanced Java A: JVM and Execution
Java	JAVA 07	Advanced Java D: Concurrency and the Memory Model
Java	JAVA 08	Advanced Java E: Performance, Diagnostics, and GC Incidents
Java	JAVA 09	Advanced Java G: Question Bank, Study Plan, and Reference
DSA	DSA 01	Time and Space Complexity for Java Interviews
DSA	DSA 02	Number Systems and Math Foundations for DSA Interviews
DSA	DSA 03	Number Systems Interview Patterns and Rapid Revision
DSA	DSA 04	Bit Manipulation in Java
DSA	DSA 05	Loop Mastery, Patterns, and Index Calculations
DSA	DSA 06	Arrays and Array Problem-Solving Patterns
DSA	DSA 07	Strings and String Problem-Solving Patterns
DSA	DSA 08	Hashing: Maps, Sets, Frequency, and Prefix State
DSA	DSA 09	Recursion and Backtracking in Java
DSA	DSA 10	Linked Lists: Pointer Reasoning and Mutation
DSA	DSA 11	Stacks, Queues, Deques, and Monotonic Patterns
DSA	DSA 12	Binary Search: Bounds, Answers, and Invariants
DSA	DSA 13	Trees, Binary Search Trees, and Tries
DSA	DSA 14	Heaps, Priority Queues, Selection, and Top-K
DSA	DSA 15	Graphs: Traversal, Ordering, Paths, and Union-Find
DSA	DSA 16	Greedy Algorithms: Recognition, Proofs, and Scheduling
DSA	DSA 17	Dynamic Programming: State, Transitions, and Optimization
System Design	SD 01	Advanced Java F: Design, Backend, Testing, and Security
System Design	SD 02	MySQL for Java Backend Interviews
System Design	SD 03	Hibernate and JPA for Java Backend Interviews
System Design	SD 04	Spring Framework for Java Engineers
System Design	SD 05	Spring Boot for Java Backend Engineers

SEGMENT	BOOK	FOCUSED LEARNING PATH
System Design	SD 06	Spring Data for Java Backend Engineers
System Design	SD 07	MongoDB for Java Backend Interviews
System Design	SD 08	Redis for Java Backend Interviews
System Design	SD 09	Apache Kafka and Spring Kafka
System Design	SD 10	Spring Ecosystem Extensions
System Design	SD 11	Spring AI for Java Engineers
System Design	SD 12	Advanced Java H: Spring Boot and REST APIs
System Design	SD 13	Advanced Java I: Persistence, SQL, JPA, and Caching
System Design	SD 14	Advanced Java J: Distributed Systems and System Design

Roadmap editions identify planned books whose chapter sets will be expanded one at a time. Existing deep-dive books remain available later in each segment, so no published material is displaced.

About the Author

Vinay Reddy Kalluri

Vinay Reddy Kalluri is a senior Java backend engineer, the creator and founding author of the Java SDE-2 Interview Preparation Series, and its Editor-in-Chief and Chief Auditor. His work spans high-throughput microservices, event-driven architecture, resilient APIs, large-scale data processing, cloud delivery, and production performance across healthcare and enterprise systems.

Editorial leadership

As Editor-in-Chief, Vinay owns the learning sequence, scope, voice, and publication decisions. As Chief Auditor, he owns the Java accuracy standard, validation evidence, attribution review, and release-readiness gate. Individual contributors retain visible credit for accepted original work through AUTHORS.md and the public Git history.

What distinguishes his approach is the combination of hands-on system design and operational ownership. He treats timeouts, retries, idempotency, dead-letter recovery, data consistency, SQL behavior, caching, and observability as part of the design rather than afterthoughts. He has also mentored engineers and led modernization, migration, and reliability work across distributed services.

Selected engineering and academic highlights

- More than eight years building Java and Spring Boot services for high-throughput healthcare and enterprise platforms.
- Designed Kafka-based event systems that have processed more than one million events per hour, with explicit idempotency, retry, recovery, and observability controls.
- M.S. in Computer Science from the University of Missouri-Kansas City (3.9/4.0) and M.S. in Software Engineering from VIT University (9.3/10), where he was a Gold Medalist.
- Independent builder of Work Visa Insights, an immigration-data intelligence platform, and Play Together, a published Android application.

Why this series exists

Vinay created this series to turn fragmented interview preparation into a deliberate progression: learn the fundamentals, run the examples, predict behavior, debug failures, explain trade-offs, and only then move into advanced engineering. The books reflect the same principle he applies to production systems: clarity before cleverness, explicit contracts, measurable behavior, and reliable foundations.

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Repository: <https://github.com/vinayreddykalluri/SDE2-Interview-Handbook>. Attribution does not imply endorsement. Java and related marks are owned by their respective holders.

Examples target Java 21 unless a section states otherwise. Validate release-specific APIs, JVM behavior, security, performance, licensing, and operational requirements before production use.

Series position: Data Structures and Algorithms - Book 17 of 17 - Study Step 17. The local table of contents and PDF bookmarks navigate within this file; the roadmap and printed filenames navigate across the complete series.

Source provenance: curated from the independent Java SDE-2 master guide plus focused original material. Original master chapter numbers are labeled explicitly when retained in cross-references.